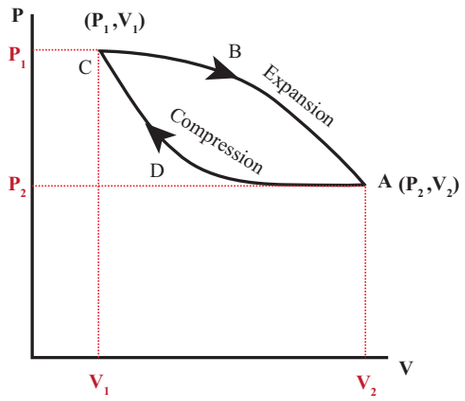


Figure 8.39 PV diagram for cyclic process

Let W_1 be the work done by the gas during expansion from volume V_1 to volume V_2. It is equal to area under the graph CBA as shown in Figure 8.40 (a) .

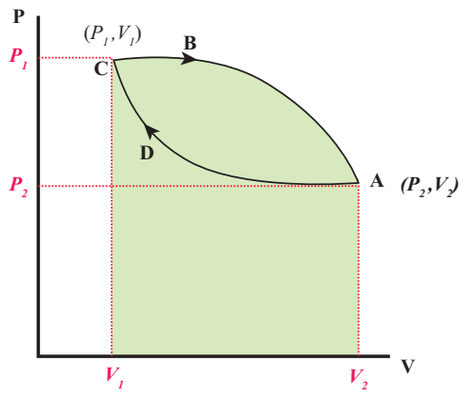


Figure 8.40 (a) W for path CBA

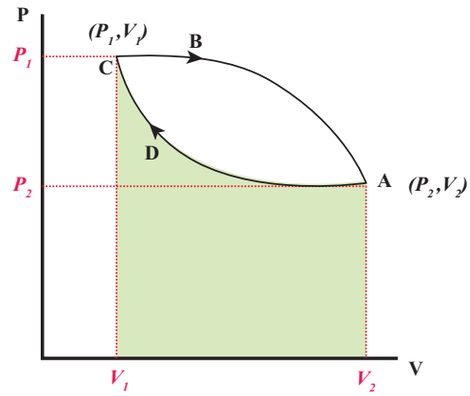


Figure 8.40 (b) W for path ADC

Let W_2 be the work done on the gas during compression from volume V_2 to volume V_1. It is equal to the area under the graph ADC as shown in Figure 8.40 (b)

The total work done in this cyclic process = W_1 - W_2 = Green shaded area inside the loop, as shown in Figure 8.41.

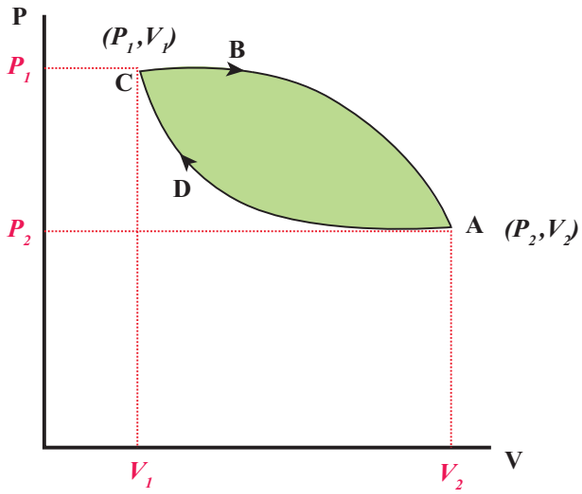


Figure 8.41 Net work done in a cyclic process

Thus the net work done during the cyclic process shown above is not zero. In general the net work done can be positive or negative. If the net work done is positive, then work done by the system is greater than the work done on the system. If the net work done is negative then the work done by the system is less than the work done on the system.

Note

Further, in a cyclic process the net work done is positive if the process goes clockwise and net work done is negative if the process goes anti-clockwise. In Figure 8.41the process goes clockwise.

UNIT-8(XI-Physics_Vol-2).indd 139 02-01-2023 15:26:09